

Document Effective Date
21/12/2021



QatarEnergy-IS RESPIRATORY PROTECTION PROCEDURE

DOC NO: IS-HSE-PRC-013

Rev: 02

Quality Reviewed By: OIH/14

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FINAL APPROVAL
OIH

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1.0 PURPOSE

The purpose of this Document is to establish a program, which will protect employees and contractors from respiratory hazards and comply with the requirements.

2.0 SCOPE AND APPLICABILITY

The Respiratory Protection Procedure applies to all employees, visitors, contractors and vendors who enter or conduct tasks in QatarEnergy-IS facilities where an air contaminant is, or may be generated.

3.0 ROLES AND RESPONSIBILITIES

RESPIRATORY PROTECTION PROGRAM ADMINISTRATOR

The Respiratory Protection Program Administrator (Program Administrator) shall be the Occupational Health Nurse who shall be responsible for:

- The selection of respirators based in accordance with their certifications.
- The review and approval of all uses of respiratory protection.
- Consulting regularly with employees required to use respirators to assess their views on program effectiveness and to identify and correct any problems.
- Conducting evaluations of the RP and to update written programs as needed,
- Ensuring breathing air analysis is performed on QatarEnergy-IS breathing air compressors,
- Maintaining records required by the program, and
- Providing supplemental respirator information and a copy of this Program to the Physician or other License Health Care Provider (PLHCP) and to the QatarEnergy-IS offshore Medics and Safety Advisors.

OFFSHORE SAFETY OFFICER / H2S TECHNICIANS / MEDICS

- Ensure that all employees (including new hires) receive appropriate training, fit testing, and medical evaluations in accordance with this Document.
- Ensure the availability of appropriate respirators and sufficient supplies (e.g., Muster BA sets, filters and disinfecting solutions)\.
- Perform a weekly inspection (including functional test) of the emergency SCBA.
- Ensure that breathing air audits are performed.
- Clean and disinfect the common tight-fitting face piece respirators used on the SAR for pigging.
- Maintain copies of the air quality test report.
- Ensure air quality tests of all incoming purchased breathing air cylinders and escape bottles, and
- Collect and have analyzed the breathing air from the platform compressors at least once a month.

SUPERVISORS

- Ensure that respirators are properly cleaned, maintained and stored.
- Enforce usage of respirators and that the proper respirator is used.

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- Ensure monthly inspection of personal respirators are documented.
- Ensure monthly inspection of the supplied air respirators are documented, and
- Ensure that engineering controls designed to control exposures are properly inspected and maintained.

EMPLOYEES

- Properly clean, inspect, maintain and store individually assigned respirators and common tight-fitting respirators,
- Use respiratory protection in accordance with the instructions and training received,
- Facial hair shall be kept in accordance with Hydrogen Sulfide Safety Procedure (IS-HSE-PRC-023), so that it does not come between the sealing surface of the face piece and the face, or interfere with valve operation, and
- Participate in medical surveillance program and fit testing program.

CONTRACTORS

QatarEnergy-IS contractors and sub-contractors shall meet the requirements listed in this Procedure at a minimum. Contractors are required to maintain sufficient documentation to demonstrate compliance.

4.0 PROCEDURAL STEPS

4.1 RESPIRATORY SELECTION

Routine Tasks

Respiratory equipment is selected to fit the potential hazards presented to operations and maintenance and input from the employees who will be using the respirators. All respirators selected, filtering face pieces and filters shall be certified by NIOSH or equivalent international Standard.

Respirators selected for routine tasks are included in the Respiratory equipment hazard assessment. A summary of the respirators selected by hazard is in Appendix A. Deviations from the specified equipment shall require the approval of the Program Administrator.

Non-Routine Tasks

For non-routine situations, such as line breaking and confined space entry, appropriate selection may be made using direct measurements. Respiratory protection required for breaking into the process or equipment and Confined Space entry will be defined in their respective Procedures.

IDLH Atmospheres

For IDLH atmospheres, the use of positive pressure through a SCBA air system with at least 30-minute capacity or supplied air respirator with auxiliary air supply is required. In addition, a stand-by person trained and equipped to provide rescue or aware of Procedures to notify trained and equipped rescuers shall be positioned outside the IDLH atmosphere and remain in visual, voice, or signal line communication with the personnel within the IDLH atmosphere.

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4.1.1 Filtering Face-piece Respirator (Dust Masks)

Filtering face piece respirators (dust masks) shall not be used for atmospheres containing contaminants for which the filtering face-piece respirator is not designed to protect against.

Filtering face-piece respirators shall be discarded when the employee notices an increase in breathing resistance, when the filter face-piece becomes discolored, when the wearer feels the need (even if that is after each use), or at least at the end of the shift.

4.1.2 Air Purifying Respirators (APRs)

Personnel who perform jobs that require the use of an APR shall have one permanently assigned. APR shall not be used to protect an employee from H₂S.

Cartridge or filter pad respirators shall change their cartridges as follows:

When employees notice a change in breathing resistance,
 When the filters become discoloured,
 When the wearer feels the need,
 At least prior to fit testing,
 At a time frame not to exceed the end of the rotation (28 days maximum), or
 Per manufacturers recommendations.
 Odor, taste, and irritation are not reliable indicators to warn of the need to change cartridges or canisters

4.1.3 Supplied Air Respirator / Self Contained Breathing Apparatus (SAR/SCBA)

SAR and SCBA provide the greatest degree of protection for the worker. Both styles of respirators use a tight fitting/sealing full facemask and function in pressure demand or other positive pressure mode.

These respirators are the only units that can be used for work in IDLH atmospheres. When using a SAR for work in an IDLH atmosphere, it shall have an egress air cylinder with sufficient capacity to permit escape to safety if the air supply is interrupted.

For work and rescue using SCBA the air cylinder shall have a capacity of thirty minutes or greater. This does not apply to the capacity of any bail out cylinder provided. SAR shall be limited to 150 feet of air hose or as approved by the manufacturer of the SAR. The regulators of both SAR and SCBA shall be flow tested annually.

Every effort should be made to avoid conducting work in IDLH environments. In those cases where personnel shall enter IDLH environments, special requirements shall be implemented including having qualified standby personnel that are properly equipped to provide assistance if required. The responsibilities of standby personnel include maintaining communication with the respirator user and rescue if needed.

4.1.4 Muster Breathing Air Respirators (BA Sets)

BA sets are SCBA with air cylinders with a minimum of air supply of fifteen minutes capacity.

All personnel enrolled in the Respiratory Protection on the platforms shall receive medical clearance, fit testing and training specific to the type of BA used and the muster sets within the past twelve months. All employees, visitors or guests shall receive training on the BA sets used for muster.

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4.1.5 Voluntary Use of Respiratory Protection

Voluntary use of respiratory protection, e.g., filtering face pieces, do not require medical evaluation or fit testing, however the employee shall be provided with the information in Appendix D.

4.2 MEDICAL EVALUATION

Personnel enrolled in the RP program who are required to wear respirators shall be evaluated to assure that they are physically and mentally able (i.e., not suffering from claustrophobia) to wear them in the work environment. Personnel shall receive an evaluation by a Physician or other Licensed Health Care Provider (PLHCP) to assess an employee's ability to use a tight-fitting facemask respirator before the employee is fit tested or required to use the respirator in the workplace by completing the form in Appendix E.

An evaluation of medical ability to wear a respirator shall be repeated every year using the form in Appendix F. Spirometry or pulmonary function tests shall be conducted every two years, unless the PLHCP or job function (e.g. Emergency Response Team member) determines more frequent exams are necessary.

The initial evaluation for all candidates for employment shall be performed during the pre-employment examination if the job position requires the use of respiratory protection. After the initial medical evaluation, employees shall receive medical evaluations at least annually, or when the following conditions occur:

- a) An employee reports signs or symptoms that are related to ability to wear a respirator,
- b) A PLHCP, supervisor, or the program administrator determines that an employee needs to be reevaluated,
- c) Information from the respirator protection program, including observations made during fit testing and program evaluation, indicates a need for employee evaluation, or
- d) A change in the workplace conditions occurs (e.g., physical work effort, protective clothing, temperatures) that may result in a substantial increase in the physiological burden placed on an employee.

Employees who voluntarily wear a filtering face-piece shall be exempted from the medical surveillance program.

All contractor employees who are required to wear respirators are required to show, upon request, proof of medical ability to wear evaluation and proper fit testing for the respiratory equipment that they will be using.

Completed medical questionnaires shall be delivered to the PLHCP. A single affirmative response in Part A, Section 2 in Appendix E requires further medical evaluation by telephone or direct contact with the PLHCP.

The Program Administrator shall provide the supplemental respirator information to the PLHCP along with a copy of this respiratory protection program prior to performing medical evaluations.

The PLHCP shall provide a medical recommendation to the Program Administrator for each employee based on Respirator Supplemental Information:

- If the employee is medically able to use a respirator;
- Any limitations on respirator use related to the medical condition of the worker;
- If necessary, a medical evaluation follow-up; and,

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- Provide the completed Appendix E to the Program Administrator so that the Program Administrator can make notifications to the employee and supervisor as required.

Communication between QatarEnergy-IS and the PLHCP shall not violate professional ethical codes, company Procedures or regulatory statutes of confidentiality. The PLHCP shall maintain confidentiality regarding medical findings and diagnoses. The PLHCP shall also provide any confidential or medical information to the employee.

4.2.1 Pulmonary Function Tests (PFT) or Spirometry

Spirometry or pulmonary function tests shall be conducted every two years, unless the PLHCP or job function determines more frequent exams are necessary. Years when an employee does not receive a spirometry, the employee shall complete Appendix F. If an employee answers “yes” to any question, that employee will need to be evaluated by the PLHCP.

4.3 FIT TESTING

All personnel and contractors who are required to use respiratory protective equipment during the course of their jobs will be fit tested annually.

QatarEnergy-IS personnel required to travel offshore as part of their responsibilities, regardless of the duration of the offshore stay, will no longer need to obtain medical approval or fit-testing for Emergency Escape Breathing Apparatus (EEBA) Set or Escape-Only Respirators, at offshore locations.

Personnel shall not use Emergency Escape BA (EEBA), for any purpose other than emergency egress to muster points or a final place of safety.

Personnel shall be fit tested using Quantitative Fit Testing (QNFT) testing protocols. Fit tests shall be conducted utilizing the same make, model and size respirator that will be used by each individual. Retesting is required when changes are made in the make, model, or size of respirator or with any change in physical condition of the worker that might affect fit (e.g. fitting of dentures, major weight loss or gain, trauma or other changes in facial structure).

4.4 PROPER RESPIRATOR USE

Respirator users shall perform a “user seal check” prior to each use of a tight-fitting respirator to verify seal and proper operation of the respirator. Refer to Appendix C for proper seal check methods.

QatarEnergy-IS shall not permit a respirator with Tight-fitting Face piece to be worn by an Employee who has facial hair that interferes with or affects the face piece seal.

QatarEnergy-IS shall not permit a respirator with Tight-fitting Face piece to be worn by an Employee with any condition that affects the face piece seal or interferes with the valve function.

Temple pieces on eyeglasses will interfere with the effective seal of full mask Tight-fitting Face pieces. Therefore, corrective and safety eyeglasses cannot be worn with full mask Tight-fitting Face pieces. The kits for mounting corrective lenses inside the face pieces without affecting the sealing surface shall be used.



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4.5 TRAINING

QatarEnergy-IS employees who are included in a Respiratory Protection will be trained on respiratory protective equipment before they are required to use respiratory protection and annually thereafter. Training shall include:

- Why the respirator is needed and how improper fit, use, or maintenance can compromise the protective effect of the respirator;
- Limitations and capabilities of the respirator;
- How to use the respirator effectively in normal situations;
- How to use the respirator effectively in emergency situations, including situations in which the respirator malfunctions;
- How to inspect, put on, adjust, and remove the respirator, visual confirmation during fit testing;
- Demonstration of positive and negative seal checks during fit testing (See Appendix C);
- Discussion of filter “breakthrough” and end-of-service change schedule if applicable;
- Discussion of the facial hair policy;

Employees who demonstrate a lack of knowledge, inappropriate use, or improper care of a respirator shall be retrained.

Employee who voluntarily use filtering face-piece respirator shall review and sign Appendix D.

Contractors are required to provide respirator training to their employees who are enrolled in a RP program which will comply with QatarEnergy-IS respiratory protection Guidelines as outlined in this Policy. This training shall be Documented and produced upon request.

4.6 RESPIRATOR CLEANING, INSPECTION, MAINTENANCE & STORAGE

Cleaning

Half face respirators shall be cleaned by the employee with a “respirator wipe” after each use. Alcohol wipes shall not be used, neither shall cleaning solutions containing oils or lanolin. When the respirator becomes soiled or where cleaning with respirator wipes becomes impractical, the respirator needs to be cleaned and disinfected by using Procedures in Appendix B. At the end of each tour, each personal half face respirator shall be disposed of or cleaned and disinfected using Procedures in Appendix B.

SCBAs for emergency and/or confined space rescue use shall be cleaned and disinfected after each use including the second stage regulator of an SCBA, but at least once a year using the Procedure in Appendix B.

BA sets shall be cleaned on a rotational basis and after each use. Muster BA sets that are found in an unsanitary condition shall be immediately returned to the BA Technician who will clean and disinfect the equipment.

Respirators used in fit testing and training will be cleaned and disinfected after each use. A spray disinfectant or respirator wipe may be used during the day between fit tests; however, respirators used to fit test more than one employee will be thoroughly cleaned and sanitized at the end of every day using the Procedure in Appendix B.



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Inspection and Maintenance

All respiratory protective equipment will be inspected prior to being placed into initial service, as well as before each use and during cleaning. Inspection Procedures must satisfy the manufacturer's requirements. All supplied air respirators shall have an annual function and posicheck test. These inspections shall be documented accordingly.

All supplied air respirators will be inspected thoroughly at least once per month. Defective equipment will be removed and replaced immediately. Each SCBA used by the emergency/fire response personnel, must be checked for proper function before and after each use. Muster BA, SCBA and emergency escape cylinders will be filled to the green zone / or equivalent.

Respirators that fail an inspection are removed from service, and are discarded, repaired or adjusted. All respirators shall be repaired only with parts from, or recommended by, the manufacturer. All repairs shall be according to manufacturer's specifications, and they will be carried out by a qualified BA Technician. All respiratory equipment must be maintained in a clean and sanitary condition at all times.

Storage

Stored respirators shall be protected from dust, sunlight, heat, extreme cold, excessive moisture, chemicals, and mechanical damage. All freshly cleaned respirator face-pieces shall be stored so that the face-piece is not deformed in any way. New filters will remain sealed until use, stored attached to the respirator after initial use, and discarded according to the cartridge change out schedule.

Spare breathing air cylinders, face pieces and regulators will be stored in good condition and away from potential physical damage.

4.7 AIR QUALITY

Breathing air produced by compressors shall be tested monthly. Preventive maintenance shall comply with manufacturer's requirements. Breathing air shall meet the following requirements (Grade D):

- Oxygen 19.5-23.5 %
- Oil (Condensed) $\leq 5 \text{ mg/m}^3$
- Carbon Monoxide $\leq 10 \text{ ppm}$
- Carbon Dioxide $\leq 1000 \text{ ppm}$
- Odor – Lack of pronounced odor
- Dew point for breathing Air Cylinders (SOBA bottles, escape packs, tube trailers, six packs, etc.) $\leq 50^\circ\text{F}$ dew point at 1 atm (63 ppm water vapor).
- Dew Point for Breathing Air Compressors $< -60^\circ\text{F}$ dew point.

Purchased Breathing Air

Each platform shall be responsible for obtaining an air quality test report to Grade D from the vendor with each shipment of breathing air indicating the air meets Grade D requirements or better. If an air quality test report is not included with the shipment, the shipment will be refused. An air quality test report shall be maintained as long as the cylinder is on the platform.

All purchased breathing air cylinders and escape bottles shall be labeled so as to be able to associate a specific cylinder with a specific air quality test report (e.g. date, lot number serial number, or attaching a copy of the report to the cylinder).

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Breathing air cylinders shall be inspected and tested per the manufacturer's requirements.

The use of portable breathing air compressors must be permitted by the platform Safety Advisor or his/her designee.

When portable breathing air cylinders are in use, an attendant, or an alarm that can be heard by the user, shall be used to notify the user to stop using the respirator when the pressure becomes low.

Each cylinder in each shipment of purchased breathing air cylinders is air quality tested to evaluate the cylinders' oxygen and carbon monoxide content. Any breathing air cylinder that fails an air quality test will be tagged "Danger - Do Not Use" and it shall be stored separately from acceptable breathing air cylinders. The HSE Manager shall be notified when a cylinder fails an air quality test and will notify the supply chain who will notify the supplier. The vendor shall conduct an investigation to determine the condition(s) that caused the unacceptable cylinder and implement corrective actions to ensure the condition(s) does not occur again. The investigation results and corrective actions shall be provided in writing to QatarEnergy-IS before any more breathing air cylinder are accepted from that vendor.

4.8 RESPIRATORY PROTECTION PROGRAM EVALUATION

The respiratory protection program shall be reviewed every three years and led by the Program Administrator. The evaluation will include the following, as a minimum:

- Audit of the respirator program operation;
- Wear acceptance survey that consults wearers on respirator fit, appropriate respirator selection for the hazards to which the wearers are exposed, and proper respirator use under the workplace conditions the wearers encounter, and proper respirator maintenance;
- Field inspection of respirator use to evaluate understanding of administrative controls, respirator storage, use, care, cleaning, disinfection, cartridge change Procedure, and compliance with facial hair policy; and
- Check of medical approval records, fit testing records and inspection records.

4.9 RECORDKEEPING

- a. Records of Employee Respiratory Protection Training. Records of employee respiratory protection training and documentation will be kept for the current year plus the past one (1) year.
Records of Employee Medical "Ability to Wear".
- b. A copy of the physician's written medical opinion will be provided to the employee and to QatarEnergy-IS upon request. Copies of the employee medical "ability to wear" shall be kept at least for the duration of employment in personnel medical files.
- c. Fit Testing
Each platform shall maintain records of respirator fit testing for the current year plus the previous year for each employee housed on the platform and in the respirator program. These records will include the following:
 - Type of respirator test performed;
 - Specific make, model, and style of respirator tested
 - Name of person tested;
 - Name of test instructor;

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- Date of test; and
 - Results of the test (overall fit factor and pass/fail indication).
- d. Inspection and Records
All respirator inspection records will be maintained for the current year plus one (1) year.

5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
HSE Training Procedure	IS-HSE-PRC-002	Internal Document	Sideward
Personal Protective Equipment Selection and Use	IS-HSE-PRC-005	Internal Document	Sideward
Hazard Communication Procedure	IS-HSE-PRC-006	Internal Document	Sideward
Contractor HSE Program	IS-HSE-PRC-009	Internal Document	Sideward
Laboratory Safety and Chemical Hygiene Procedure	IS-HSE-PRC-014	Internal Document	Sideward
Hydrogen Sulfide Safety Procedure	IS-HSE-PRC-023	Internal Document	Sideward
Industrial Hygiene Protection Procedure	IS-HSE-PRC-024	Internal Document	Sideward
Confined Space Entry Procedure	IS-FOP-PRC-206	Internal Document	Sideward

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6.0 DEFINITIONS

Term	Abbreviation
Dew Point	Temperature at which the air is saturated with moisture.
Fit Testing	An assessment of the adequacy of respirator fit by numerically measuring the amount of leakage into the respirator.
Immediate Dangerous to Life and Health (IDLH)	Atmosphere that poses an immediate threat to life, would cause irreversible adverse health effects, or would impair an individual's ability to escape from a dangerous atmosphere.
Physician or Licensed Health Care Provider (PLHCP)	Individual whose legally permitted scope of practice (i.e., license, registration, or certification) allows him or her to independently provide, or be delegated the responsibility to provide, some or all of the health care services.
Self-Contained Breathing Apparatus (SCBA)	Means an atmosphere-supplying respirator for which the source of breathing air is designed to be carried by the user.
Supplied-Air Respirator (SAR) or airline respirator	Atmosphere-supplying respirator for which the source of breathing air is not designed to be carried by the user.
Tight-fitting face piece	A respiratory inlet covering that forms a complete seal with the face

APPENDIX

APPENDIX A - QatarEnergy-IS RESPIRATORY HAZARDS

Workplace Health/Hygiene Hazard	Common Source / Area	Physical State	Health/Hygiene & Safety Effects	TLV*	Control Methods
Asbestos	Electrical switch room	Dust	Lung irritant, carcinogenic	0.01 fibers/cc	Encapsulated, Proceduralized when disturbed
Benzene	Glycol unit	Fumes	Toxic, flammable	10 ppm	Full face mask
Glycol	Gas dehydration	Fumes	Irritant, non-respiratory	50 ppm	Procedural
H ₂ S	Process, well heads	Gas	Toxic, flammable	10 ppm	Full face mask or PARAT mask ****

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Lead	Blasting and coating	Dust	Toxic	0.1 mg/m ³	Procedural
Silica Quartz	Blasting and coating	Dust	Lung irritant	0.5 mg/m ³	NIOSH approved respirator, e.g. P-100 filters
Solvents, Thinners: e.g. Toluene	Coating and painting Laboratory, process	Fumes	Irritant	100 ppm**	NIOSH approved respirator, e.g. P-100 filters
NORM	Process equipment	Dust	Body tissue damage, carcinogenic	50mSv effective dose	Procedural, permit control, NIOSH approved respirator or full-face mask
Welding Fumes, e.g., Cadmium	Welding operation in enclosed spaces	Fumes	Irritant, Carcinogenic	0.005mg/m ³ ***	Air-hood, MSDS for other specific metals

Respiratory protection shall be used when TLV levels are exceeded.

* Threshold Limit Value: ACGIH latest edition.

** Paints and thinners use many solvents; the TLV listed is for toluene.

*** There are many fumes emitted during a typical welding operation with cadmium being the primary fume; thus, the TLV listed is for cadmium.

**** PARAT masks may be used to reach a full-face muster BA set only.

APPENDIX B - RESPIRATOR CLEANING PROCEDURES

This Procedure is based on advice by the manufacturer of the respirators. Respirator wipes may be used for the limited of the respirators. At least once a year, prior to fit testing, all personal issued half face respirators shall be cleaned as outlined in this Appendix.

Procedures for Cleaning Respirators

- Remove filters or regulators. Disassemble face pieces by removing speaking diaphragms, demand and pressure-demand valve assemblies, hoses, or any components recommended by the manufacturer. Discard or repair any defective parts.
- Wash components in warm (43 deg. C [110 deg. F] maximum) water with a mild detergent or with a cleaner recommended by the manufacturer. A stiff bristle (not wire) brush may be used to facilitate the removal of dirt. Do not use cleaners with lanolin or oils.

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- c. Rinse components thoroughly in clean, warm (43 deg. C [110 deg. F] maximum), preferably running water. Drain.
- d. When the cleaner used does not contain a disinfecting agent, respirator components should be immersed for two minutes in one of the following:
 - One milliliter of household bleach to one liter of water at 43 deg. C (110 deg. F); or,
 - Other commercially available cleansers of equivalent disinfectant quality when used as directed, if their use is recommended or approved by the respirator manufacturer.
- e. Rinse components thoroughly in clean, warm (43 deg. C [110 deg. F] maximum), preferably running water. Drain. The importance of thorough rinsing cannot be overemphasized. Detergents or disinfectants that dry on face-pieces may result in dermatitis. In addition, some disinfectants may cause deterioration of rubber or corrosion of metal parts if not completely removed.
- f. Components should be hand-dried with a clean lint-free cloth or air-dried.
- g. Reassemble face-piece, replacing filters, where necessary.
- h. Test the respirator to ensure that all components work properly.

APPENDIX C - USER SEAL CHECKS

User Seal Checks are to be performed each time the respirator wearer dons (puts on) the respirator. The seal checks are required as part of the fit testing Procedure; however, a User Seal Check is NOT the same as the Respirator Fit Test.

Seal Check for Negative Pressure Respirators

There are two common Procedures a worker shall perform to check a face-piece seal each time a negative pressure respirator (including full face, half-face and quarter face respirator) is put on. At least one of the following two Procedures must be performed.

- Positive pressure check
- Negative pressure check

The respirator manufacturer's recommended Procedures for performing a user seal check may be used instead of the positive and/or negative pressure check Procedures provided that the you can demonstrates that the manufacturer's Procedures are equally effective.

Positive Pressure Check

To perform a positive pressure check, follow these steps:

1. Cover the exhalation valve of the respirator.
2. Exhale gently for about 10 seconds. Don't exhale too hard or push the mask into the face or the check will be inaccurate.

If the respirator fits, a slight pressure should build up inside it. If air leaks out, the respirator does not fit properly, and the seal is inadequate. Figure below illustrates a positive pressure check.



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Negative Pressure Check

To perform a negative pressure check, follow these steps:

1. Cover the filter openings with the palms of hands.
2. Inhale gently and hold a breath for about 10 seconds. Don't push the respirator into the face too hard or the check will be inaccurate.

If the face-piece fits correctly, it should collapse slightly inward. If the face-piece does not fit correctly, it will not collapse, and an air leak will be felt. This test is done on SCBA by covering the hose with the hand and inhaling.



Negative pressure fit check

APPENDIX D - INFORMATION FOR EMPLOYEES USING RESPIRATORS WHEN NOT REQUIRED UNDER THE PROGRAM

Respirators are an effective method of protection against designated hazards when properly selected and worn. Respirator use is encouraged, even when exposures are below the exposure limit, to provide an additional level of comfort and protection for workers. However, if a respirator is used improperly or not kept clean, the respirator itself can become a hazard to the worker.

Sometimes, workers may wear respirators to avoid exposures to hazards, even if the amount of hazardous substance does not exceed the limits set. If your employer provides respirators for your voluntary use, or if you provide your own respirator, you need to take certain precautions to be sure that the respirator itself does not present a hazard.

You should do the following:

1. Read and heed all instructions provided by the manufacturer on use, maintenance, cleaning and care, and warnings regarding the respirators limitations.
2. Choose respirators certified for use to protect against the contaminant of concern. NIOSH, the National Institute for Occupational Safety and Health of the U.S. Department of Health and Human Services, certifies respirators. A label or statement of certification should appear on the respirator or respirator packaging. It will tell you what the respirator is designed for and how much it will protect you.
3. Do not wear your respirator into atmospheres containing contaminants for which your respirator is not designed to protect against. For example, a

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- respirator designed to filter dust particles will not protect you against gases, vapors, or very small solid particles of fumes or smoke.
4. Keep track of your respirator so that you do not mistakenly use someone else's respirator.

I have read and understood the “Appendix 4” information regarding voluntary use of a respirator.

Employee (Signature)

Date

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APPENDIX E - RESPIRATOR MEDICAL EVALUATION QUESTIONNAIRE

Employee Name:		Male / Female
Employee Number:	Date of Birth:	Sex (circle one)
Job Title:		
Location:	Supervisor's Name:	
Height:	Weight:	
Phone Number	Best Time to Call	

Section A

Questions	Yes	No
1. Can you read?	☐	☐
2. Has your employer told you how to contact the health care professional who will review this questionnaire?	☐	☐
3. Would you like to talk to the health care professional who will review this?	☐	☐

4. Check the work level that best describes your job:

- ☐ Light Work sitting, drafting, performing light assembly work
- ☐ Medium Work driving a truck in urban traffic, standing while drilling or performing assembly work, transfer 35 lbs. trunk level, pushing a load of 100 lbs. with a wheelbarrow
- ☐ Heavy Work lifting a heavy load about 50 lbs. from floor to your waist or shoulder, working on a loading dock

5. Check the type of respirator you will use and indicate frequency of use:

Respirator Type		Less than 2 hours per week	Less than 5 hours per week	Over 4 hours per week	Emergency escape only
☐	Disposable respirator (filter mask, non-cartridge type (Non-Powered respirators)				
☐	Other type (for example, half or full-face piece type, powered-air purifying, supplied-air, self-contained breathing apparatus).				

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	Yes	No
6. Will you be wearing protective clothing and/or equipment when you are using your respirator?	☐	☐

If “yes” describe this protective clothing and/or equipment

7. Describe any special or hazardous conditions you might encounter when you are using your respirator(s)

Questions	Yes	No
8. Will you be working under hot conditions?	☐	☐
9. Will you be working under humid conditions?	☐	☐

List all current medications:

Section B

Question	Yes	No
10. Have you ever worn a respirator?	☐	☐

If “yes” what type (s): _____

11. If you have used a respirator, have you ever had any of the following problems?

Yes	No		Yes	No	
☐	☐	Eye irritation	☐	☐	Anxiety
☐	☐	Skin allergies or rashes	☐	☐	General weakness or fatigue
☐	☐	Any other problem that interferes with your use of a respirator			

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12. Have you ever had any of the following problems?

Yes	No		Yes	No	
☐	☐	Seizures	☐	☐	Swelling in your legs/feet (not caused by walking)
☐	☐	Diabetes (sugar disease)	☐	☐	High blood pressure
☐	☐	Allergic reactions (interfere w/ breathing)	☐	☐	Heart failure
☐	☐	Claustrophobia (fear of closed-in places)	☐	☐	Heart arrhythmia (heart beating irregularly)
☐	☐	Trouble smelling odors	☐	☐	Heartburn or indigestion that is not related to eating
☐	☐	Asbestosis			
☐	☐	Asthma			
☐	☐	Chronic bronchitis	☐	☐	In the past 2 years, have you noticed your heart skipping or missing a beat
☐	☐	Emphysema	☐	☐	Pain or tightness in your chest
☐	☐	Pneumonia	☐	☐	Pain or tightness in your chest during physical activity
☐	☐	Tuberculosis	☐	☐	Pain or tightness in your chest interferes with your job
☐	☐	Silicosis	☐	☐	Other problems that you think may be related to heart problems
☐	☐	Angina	☐	☐	Other heart problems you have been told about
☐	☐	Pneumothorax (collapsed lung)			
☐	☐	Lung cancer	☐	☐	Other problems that you think may be related to heart problems

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☐	☐	Broken ribs	☐	☐	Other heart problems you have been told about
☐	☐	Chest injuries or surgeries			
☐	☐	Heart attack			
☐	☐	Stroke			

13. Do you currently have any of the following symptoms?

Yes	No		Yes	No	
☐	☐	Shortness of breath	☐	☐	Wheezing
☐	☐	Shortness of breath when washing or dressing	☐	☐	Coughing that produces phlegm (through sputum)
☐	☐	Shortness of breath that interferes with your job	☐	☐	Coughing up blood in the last month
☐	☐	Coughing that wakes you early in the morning	☐	☐	Coughing that occurs mostly when you are lying down
☐	☐	Chest pain when you breathe deeply	☐	☐	Any other symptoms that you think be related to lung problems
☐	☐	Shortness of breath when walking fast on level ground or walking up a slight hill or incline	☐	☐	Wheezing that interferes with your breath
☐	☐	Shortness of breath when walking with other people at an ordinary pace on level ground			
☐	☐	Have to stop for breath when walking at your own pace on level ground			

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14. Do you currently take medication for any of the following problems?

Yes	No		Yes	No	
☐	☐	Breathing or Lung Problems	☐	☐	Seizures
☐	☐	Heart Trouble Blood Pressure	☐	☐	
☐	☐	List all medications you are taking	☐	☐	

List all medications you are taking:

Question	Yes	No
15. Do you currently smoke tobacco, or have you smoked tobacco in the last month?	☐	☐
_____ Packs of cigarettes per day		_____ Year Quit
_____ Cigars per week		_____ Number of years smoked
_____ Pouches or pipe tobacco per week		
Explain all “yes” answers to questions 10 through 15		

Section C

Questions	Yes	No
16. Have you ever had a back injury?	☐	☐
17. Have you ever lost vision in either eye (Temporarily or permanently)?	☐	☐

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18. Do you currently have any of the following vision problems?

Yes	No		Yes	No	
☐	☐	Wear Contact Lenses	☐	☐	Color Blind
☐	☐	Wear Glasses	☐	☐	Any other eye or vision problem

Yes No

19. Ever had an injury to your ears, including a broken eardrum (perforated)?

☐ ☐

20. Do you currently have any of the following hearing problems?

Yes	No		Yes	No	
☐	☐	Difficulty hearing	☐	☐	Any other hearing or ear problem
☐	☐	Wear a hearing aid	☐	☐	

21. Do you currently have any of the following musculoskeletal problems?

Yes	No		Yes	No	
☐	☐	Weakness in any of your arms, hands, legs, feet	☐	☐	Difficulty bending at your knees
☐	☐	Back pain	☐	☐	Difficulty squatting to the ground
☐	☐	Difficulty fully moving your arms and legs	☐	☐	Difficulty fully moving your head side to side
☐	☐	Difficulty fully moving your head up or down	☐	☐	Climbing a flight of stairs/ladder carrying 25 lbs. plus
☐	☐	Pain or stiffness when you lean forward or backward	☐	☐	Any other muscle or skeletal problem that interfere with using a respirator

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Explain all “yes” answers to questions 17 through 22

Employee Signature: _____

Date: _____

Health Professional Review

Print

Date

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APPENDIX F - QUESTIONNAIRE PRIOR TO ANNUAL FIT TESTING

Name: _____ DOB: _____

Department: _____ Job: _____

Please answer YES or NO to each question

Questions	Yes	No
1. Have you developed any medical problems or symptoms that may limit your ability to use a respirator?	☐	☐
2. Have you been told by a health care professional, your supervisor, or the respirator program administrator that you should be medically reevaluated?	☐	☐
3. Has there been a change in workplace conditions, e.g., physical work effort, protective clothing, temperature, that has resulted in a substantial increase in the physical burden on you?	☐	☐

It is your responsibility to report to your supervisor any change in health status that may affect your ability to use a respirator.

Employee Signature: _____ Date: _____

- ☐ No medical follow-up is necessary; proceed with fit test.
- ☐ Medical evaluation is indicated; do not proceed with fit test.

Reviewer Signature: _____ Date: _____

APPENDIX G – REVISION HISTORY LOG

Item Revised	Revision Description	Page No.
		6
Remarks:		

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APPENDIX H – LIST OF ABBREVIATIONS

Term	Abbreviation
APR	Air Purifying Respirators
EEBA	Emergency Escape Breathing Apparatus
IDLH	Immediate Dangerous to Life and Health
PLHCP	Physician or Licensed Health Care Provider
PLHCP	Physician Or Other Licensed Health Care Provider
SCBA	Self-Contained Breathing Apparatus
SAR	Supplied-Air Respirator